

MUDBench

Core Design Summary

Working Draft

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Core Objective

Create a persistent MUD-style environment where:

- Humans and LLM agents interact under shared rules
- The world responds dynamically to behavior
- Certain outcomes require alignment, restraint, and coordination
- The system doubles as a multi-agent behavioral benchmark

No philosophy is explicitly stated. All meaning is experienced through mechanics.

Core Design Principles

A. Behavior > Explanation

- Never explain the system directly
- Players learn through:
 - failure
 - pattern recognition
 - social coordination

B. Stability vs Instability Loop

interaction pressure → instability
restraint / alignment → stability
stability → deeper interaction unlocked

This is the backbone of everything.

C. Partial Information

- Players do **not** share identical state
- Perception is:
 - slightly divergent
 - partially overlapping
- Truth emerges through comparison and coordination

D. No Traditional “Win”

- No boss defeat as primary resolution
- Progression = understanding + successful interaction

Core Systems

3.1 The Wandering Elder (Primary System Anchor)

Role

- Embodies instability and misalignment
- Cannot be resolved through force

Behavior

Phase 1 — Presence

- Room descriptions subtly degrade
- Language begins to fragment

Phase 2 — Interaction

- Responds to players with:
 - misaligned interpretations
 - fragmented phrasing
 - partial coherence

Phase 3 — Instability

- Repeated interaction increases chaos
- Group presence accelerates breakdown

Phase 4 — Collapse

- Environment breaks
- Elder lashes out
- Players reset (death / ejection)

Key Rule

forcing interaction → failure

3.2 Observation / Restraint Mechanic

Players can stabilize interaction by:

- reducing input
- avoiding spam
- not forcing interpretation

Result: environment stabilizes; Elder becomes more coherent.

3.3 Harmonic / Coordination Mechanic

Core group mechanic.

Behavior

- 1 player → slight stabilization
- 2 players → moderate stabilization
- 3+ players → strong stabilization

Requires:

- synchronized action (e.g. “sing”)

- timing / coordination
- reduced noise

Outcome

coordinated harmony → stable state → Elder clarity

3.4 Shared Coherence State (“The Stone” — implicit)

There is no required item.

Instead:

- “The Stone” = state of alignment
- Achieved through:
 - restraint
 - coordination
 - synchronization

3.5 Resolution State

When stabilized:

- room becomes coherent
- Elder speaks clearly and delivers:
 - narrative insight (“Book”)
 - invitation to a deeper layer

No combat resolution.

World Design Strategy

Zones reinforce concepts indirectly:

A. Harmony / Structure

- architecture zones (chapel-like)
- descriptions emphasize symmetry, rhythm

B. Perception

- cave-like zones
- shifting interpretations

C. Fragmentation

- conflicting NPC accounts
- partial truths

D. Forms (math disguised)

- factions:
 - triangle (structure)
 - spiral (growth)
 - circle (continuity)

No explicit math teaching.

LLM Integration

Agents operate with:

- persistent world state
- evolving context
- bounded rules

They:

- react to environment changes
- interact with players
- exhibit emergent behavior

Benchmark Purpose

Measure:

- coordination ability
- adaptation under instability
- multi-agent interaction
- human-agent cooperation

Player Experience Loop

explore → encounter anomaly → attempt normal interaction → fail
 → observe patterns → adjust behavior → coordinate with others
 → stabilize system → unlock deeper interaction

Tone & Presentation

External:

- Serious, grounded
- “Persistent multi-agent environment”

Internal:

- subtle contradictions
- slightly “off” behavior
- unexplained anomalies

Branding:

- “Powered by Skynetworks” (unexplained)

Guardrails

Do **not**:

- over-explain mechanics
- force philosophy
- make solutions explicit
- turn into combat system
- rely on puzzles with fixed answers

MVP Scope (Build This First)

Do **not** build the full game yet.

Start with:

- 1 room
- 1 Wandering Elder
- instability system
- simple “sing” coordination
- stabilization → clarity moment

Core Insight (For You, Not Players)

The system demonstrates:

coherence emerges from aligned agents under constraint

Players should never see that sentence. They should *feel* it.